

# Multi-associative property of Hopfield networks based on relaxed spins

Ali Zahid\*

Science School, ?????????

Mikhail Erementchouk<sup>†</sup> and Pinaki Mazumder<sup>‡</sup>

Department of Electrical Engineering and Computer Science,  
University of Michigan, Ann Arbor, MI 48109 USA

## I. INTRODUCTION

## II. BASIC PROPERTIES OF THE $V_2$ MODEL AND HOPFIELD NETWORKS

A Hopfield network stores and retrieves information through the dynamics of a network of pair-wise coupled classical spins (binary variables taking values  $\pm 1$ ). The general properties of Hopfield networks are overviewed in many publications [? ]. Therefore, here, we only remind several features playing the key role in the following. Hopfield networks encode information in weights of coupling between spins.

### A. Hopfield networks

We start with the case of a network carrying only one image  $\xi = \{\xi_1, \xi_2, \dots, \xi_N\}$ , where  $\xi_n \in \{-1, 1\}$  represent the image pixels, and  $N$  is the total number of pixels. Throughout the paper, we adopt the convention that set and unset pixels are represented by  $\xi_n = +1$  and  $\xi_n = -1$ , respectively. The enumeration of pixels is supplemented with the correspondence between the pixel number and its position in the image.

Let the network be in spin state  $\sigma = (\sigma_1, \dots, \sigma_N)$ , where  $\sigma_n \in \{-1, 1\}$ , then the overlap between the state and the encoded image is quantified by the scalar product of  $\xi$  and  $\sigma$

$$Q_\xi(\sigma) \equiv (\xi, \sigma) = \sum_m \xi_m \sigma_m. \quad (1)$$

The overlap reaches maximum if and only if  $\sigma = \xi$ , that is if the network spin state reproduces the encoded image. In view of further extension to multiple images, reproducing the encoded image is associated with the maximum of objective function

$$G(\sigma) = \frac{1}{4} Q_\xi^2(\sigma) = \frac{N^2}{4} + \frac{1}{4} \sum_{m \neq n} \xi_m \xi_n \sigma_m \sigma_n. \quad (2)$$

The objective function  $G(\sigma)$  is strictly quadratic in spin variables and, therefore, is symmetric with respect to the global spin inversion  $G(\sigma) = G(-\sigma)$ . Thus, the objective function is maximized by both,  $\xi$  and  $-\xi$ . In the associative-memory setting, this degeneracy is lifted by the input pattern, which fixes the sign of the desired image.

### \*\*\* Negative pattern

The construction is generalized to encoding multiple,  $P > 1$ , images,  $\mathcal{P} = \{\xi^{(1)}, \dots, \xi^{(P)}\}$ , assuming that no two images are identical up to global inversion,  $|\left(\xi^{(\mu)}, \xi^{(\nu)}\right)| < N$  for  $\mu \neq \nu$ . In anticipation of the relaxed spin-based realization of the Hopfield network, we require that setting the spin configuration to any encoded image results in the same value of the objective function. This is achieved by using a modified Hebbian rule

$$G(\sigma) = \frac{1}{4} \sum_{\mu=1}^P \lambda_\mu Q_\mu^2(\sigma), \quad (3)$$

where  $Q_\mu(\sigma) = \left(\xi^{(\mu)}, \sigma\right)$ , and  $\lambda_\mu$  are normalization coefficients determined by the system of equations

$$\sum_{\mu} \lambda_\mu \left(\xi^{(\nu)}, \xi^{(\mu)}\right)^2 = N^2, \quad (4)$$

for all  $\nu = 1, \dots, P$ . We note that system (4) is trivial when the network stores orthogonal images,  $\left(\xi^{(\mu)}, \xi^{(\nu)}\right) = N\delta_{\mu,\nu}$ , with the solution  $\lambda_\mu = 1$ . At the same time, as we will see, the case of orthogonal images, while nontrivial, is the least interesting.

The objective function  $G(\sigma)$  defined by Eq. (3) possesses the same symmetry with respect to the global inversion of spins as for the single image case. Another important symmetry is a symmetry with respect to component-wise “gauge” transformation on the binary hypercube. For binary vectors  $\xi, \eta \in \{-1, 1\}^N$ , we define  $(\xi \odot \eta)_m = \xi_m \eta_m$ . This operation is commutative and associative, has identity  $\mathbf{e} \equiv (1, \dots, 1)$ , satisfying  $\mathbf{e} \odot \xi = \xi$ , and has the property  $\xi \odot \xi = \mathbf{e}$ . It is easy to see that this transformation preserves the scalar product,  $(\eta \odot \xi, \eta \odot \zeta) = (\xi, \zeta)$ , for any  $\zeta \in \{-1, 1\}^N$ , and, therefore, preserves the objective function, while modifying the set of stored images.

Thus, multiplying each vector in the set  $\mathcal{P}$ , say, by  $\xi^{(1)}$ , we obtain a new set  $\tilde{\mathcal{P}} = \{\tilde{\xi}^{(1)}, \dots, \tilde{\xi}^{(P)}\}$ , where

\* zahida@bxscience.edu

† merement@gmail.com

‡ pinakimazum@gmail.com

$\tilde{\xi}^{(\mu)} = \xi^{(1)} \odot \xi^{(\mu)}$  and  $\tilde{\xi}^{(1)} = \mathbf{e}$ . The original images are restored by multiplying by  $\xi^{(1)}$  again. This allows us to assume without loss of generality that any chosen image in  $\mathcal{P}$  has all pixels set. Since the transformation is involutive, the spin configuration in the original gauge is obtained by the same transformation,  $\sigma \rightarrow \xi^{(1)} \odot \sigma$ .

To relate the operation of a Hopfield network recovering a stored image to the progression of an Ising machine, it is constructive to note the combinatorial meaning of  $G(\sigma)$ . To this end, we regard the term  $\sigma_m \sigma_n$  in Eq. (2) and the expansion of Eq. (3) as stemming from the expansion of the cut indicator function  $\chi_{m,n} = (1 - \sigma_m \sigma_n)/2$  in the network graph  $\mathcal{G}$  with the spin configuration determining the distribution of the graph nodes between two partitions. This yields the relationship between  $G(\sigma)$  and the weighted cut of the network graph  $\mathcal{G}$

$$\begin{aligned} C_{\mathcal{G}}(\sigma) &= \frac{1}{2} \sum_{\mu, m, n} \lambda_{\mu} \xi_m^{(\mu)} \xi_n^{(\mu)} \chi_{m,n} \\ &= \frac{1}{4} \sum_{\mu} \lambda_{\mu} \left( \mathbf{e}, \xi^{(\mu)} \right)^2 - G(\sigma). \end{aligned} \quad (5)$$

Thus, recovering image  $\xi$  is equivalent, up to an additive constant, to *minimizing* a weighted signed cut of a complete graph  $\mathcal{K}_N$  with the weighted adjacency matrix  $\mathbf{A}_{\mathcal{P}} = \sum_{\mu} \lambda_{\mu} \left( \xi^{(\mu)} \xi^{(\mu)T} - \mathbf{I} \right)$ , or, alternatively, to *maximizing* the cut for  $-\mathbf{A}_{\mathcal{P}}$ . The identity matrix  $\mathbf{I}$  ensures that the diagonal elements of the adjacency matrix are zero.

### B. $V_2$ model

The relaxed spin realization of the Hopfield network is obtained by assigning relaxed spins to the network's nodes. A relaxed spin is a pair  $\psi_n = (\sigma_n, X_n)$ , where  $\sigma_n \in \{-1, 1\}$  and  $X_n \in (-1, 1]$  are the discrete and continuous components of the relaxed spin, respectively. Thus, the state of the network is characterized by the relaxed spin configuration  $\psi = (\sigma, \mathbf{X})$ .

#### \*\*\* Phase-space

$$C_{V_2}(\psi) = C_{\mathcal{G}}(\sigma) + \Delta C(\psi), \quad (6)$$

where

$$\Delta C(\psi) = \frac{1}{4} \sum_{m, n} A_{m, n} \sigma_m \sigma_n |X_m - X_n| \quad (7)$$

#### \*\*\* Equations of motion

### III. MULTI-STATE STRUCTURE

$$\begin{array}{ccccccc} 1 & > & X^{(1)} & > & X^{(2)} & > & X^{(3)} & > & \dots \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \emptyset & & \mathcal{V}^{(1)} & & \mathcal{V}^{(2)} & & \mathcal{V}^{(3)} & & \dots \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \emptyset & \subset & \mathcal{J}^{(1)} & \subset & \mathcal{J}^{(2)} & \subset & \mathcal{J}^{(3)} & \subset & \dots \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \xi^{(i_0)} & \prec & \xi^{(i_1)} & \prec & \xi^{(i_2)} & \prec & \xi^{(i_3)} & \prec & \dots \end{array} \quad (8)$$

### IV. TERMINAL STATES: SINGLE IMAGE

As we will discuss in detail below, the ability to contain multiple inference outcomes is an inherent property of the phase space of relaxed spins. However, the convergence to respective states is the specific feature of the  $V_2$  model. We start an analysis of the related class of problems from considering a simple case of the network storing only one image (up to the global spin inversion). While this case is not representative of the multiple-output property, it demonstrates main features of the terminal states of the  $V_2$  model.

As discussed above, we can assume that the stored image has all pixels set. Then, the problem of recovering the  $N$ -pixel image is equivalent to finding the minimum cut of a complete graph  $\mathcal{K}_N$  with edges having unit weight. This problem has a trivial solution with all nodes assigned to one partition, so that there are no cut edges. From the relaxed spins perspective, this corresponds to all relaxed spins grouping into a single cluster with the same values of the discrete and continuous components.

The dynamics of the  $V_2$  progressing towards the minimum cut is determined by the simple “ferromagnetic” rule: when regarding relaxed spins arranged on the interval  $[-1, 1]$ , co-oriented spins attract and counter-oriented spins repel each other. It follows then that the single-cluster state corresponding to the minimum cut is a stable equilibrium: any displacement of the relaxed spins within the interval  $[-1, 1]$  will lead to collapsing the relaxed spins back to a single cluster.

However, the  $V_2$  model also admits other equilibria. We consider those that are stable with respect to sufficiently small perturbations. Let  $(\sigma(0), \mathbf{X}(0))$  be such a metastable state, then there exists sufficiently small  $\lambda > 0$  such that starting from  $(\sigma(0), \mathbf{X}(0) + \lambda \mathbf{Y})$ , the network terminates in state  $(\sigma(0), \mathbf{X}(\infty))$ , with the same discrete component of the relaxed spins, for any  $\mathbf{Y}$ .

Let  $(\sigma(0), \mathbf{X}(0))$  have the form of  $S$  clusters  $\mathcal{V}^{(1)}, \dots, \mathcal{V}^{(S)}$ , where  $\mathcal{V}^{(p)} = \{u \in \mathcal{V} : X_u = X^{(p)}\}$ , enumerated in such a way that  $X^{(p)} > X^{(q)}$  for  $p > q$ . Then, using the similar approach as in [?], one can show that  $S$  is odd, each cluster contains the same number,  $|\mathcal{V}^{(p)}| = N/S$ , of co-oriented relaxed spins, and the neighboring clusters have relaxed spins with the discrete components with opposite signs.

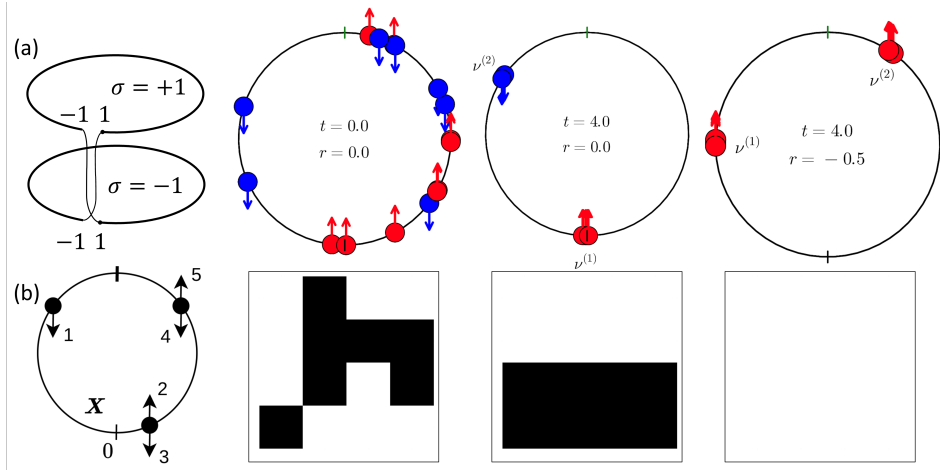


FIG. 1. Evolution of a relaxed-spin network. The green dash at the top marks the  $[-1, 1]$  boundary, the black dash at the bottom is  $X = 0$ . Spin up (red) and down (blue) correspond to  $\sigma = +1$  and  $\sigma = -1$ , respectively. (a) The initial state of a relaxed spin system of  $N = 16$ . (b) The spin system after evolving to  $t = 4.0$  with 3000 steps. Two distinct clusters  $\mathcal{V}^{(1)}$  and  $\mathcal{V}^{(2)}$  formed. Note the existence of clusters  $\nu^{(1)}$  and  $\nu^{(2)}$ . These will then be rotated to obtain the multiple images or associations. (c) State from (b) was rotated by shifting all nodes by  $-0.5$ . Since  $\nu^{(2)}$  crossed the  $[-1, 1]$  boundary, all its spins flip. This yields a new associations while still keeping cut the same.

To show this, we associate charge  $Q^{(p)} = \sum_{u \in \mathcal{V}^{(p)}} \sigma_u$  with the  $p$ -th cluster. Next, we assume that the perturbation  $\mathbf{X}(0) \rightarrow \mathbf{X}(0) + \lambda \mathbf{Y}$  spreads each cluster over non-overlapping intervals. Then, the equation of motion of a relaxed spin  $(\sigma_u, X_u)$  in the perturbed cluster  $\mathcal{V}^{(p)}$  can be written in the form

$$\dot{X}_u = \dot{X}_u^{(p)} + \sigma_u \epsilon(\bar{p}), \quad (9)$$

where  $\dot{X}_u^{(p)}$  is the contribution due to coupling to other relaxed spins in  $\mathcal{V}^{(p)}$ , and the last term describes the effect of other clusters with  $\epsilon(\bar{p}) = \sum_{q > p} Q^{(q)} - \sum_{q < p} Q^{(q)}$ . Adding up Eqs. (7) over  $\mathcal{V}^{(p)}$  and taking into account the asymmetry of the equations of motion, we obtain

$$\dot{X}^{(p)} = Q^{(p)} \epsilon(\bar{p}), \quad (10)$$

where  $\dot{X}^{(p)} = \sum_{u \in \mathcal{V}^{(p)}} \dot{X}_u$  is the velocity of the “center of mass” of the  $p$ -th cluster. We notice that the right-hand side of Eq. (8) does not depend on the perturbation as long as it does not lead to overlapping clusters. Thus, in equilibrium one has either  $Q^{(p)} = 0$  or  $\epsilon(\bar{p}) = 0$  (or both). However, owing to the ferromagnetic coupling, clusters with  $Q^{(p)} = 0$ , or even simply containing relaxed spins with opposite discrete components, are inherently unstable: there are small perturbations leading to dispersing such a cluster and, eventually, to changing  $\sigma$ . Thus, we must have  $|Q^{(p)}| = |\mathcal{V}^{(p)}|$  and  $\epsilon(\bar{p}) = 0$  for all  $p$ .

Considering the condition  $\epsilon(\bar{p}) = 0$  consequently for  $p = 1, 2, \dots, S$ , one finds that  $Q^{(p)} = -Q^{(p+1)}$ , for  $1 \leq p < S$ , and  $Q^{(1)} = Q^{(S)}$  [? ]. Thus,  $S$  must be an

odd divisor of  $N$ , and each cluster must have the same number of the relaxed spins:  $|\mathcal{V}^{(p)}| = Q$  and  $N = QS$ .

Without going into a detailed analysis of emergence of structured sub-optimal states from generic initial conditions, we limit ourselves to considering images with the

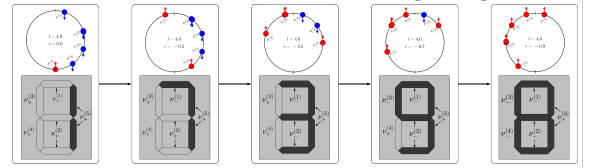


FIG. 2.

number of pixels without odd divisors (powers of two), for which sub-optimal states are excluded.

## V. TERMINAL STATES: MULTIPLE IMAGES

### A. Chains of images

### B. Stability of chains

### C. Orthogonal images

## VI. CONCLUSION

## ACKNOWLEDGMENTS

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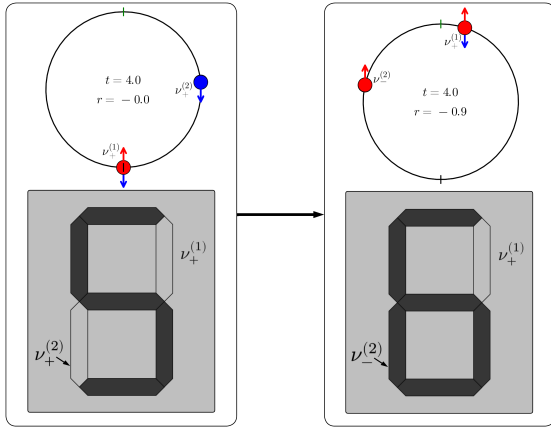


FIG. 3.

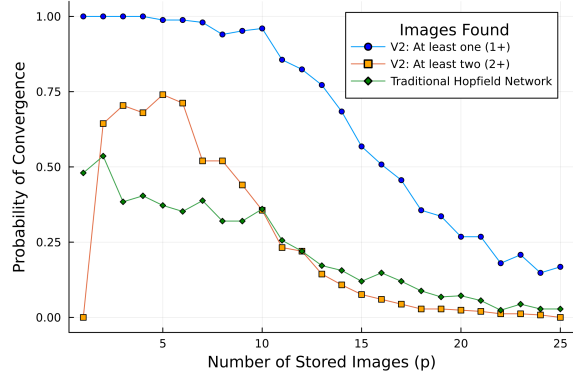


FIG. 4. Condition II (mutually orthogonal images) with  $16 \times 16$  images with 250 iterations per data point. Blue is the probability of converging to a valid image for the  $V_2$  model. Orange is the probability of converging to multiple images for the  $V_2$  model. Green is the probability of converging to a valid image for traditional Hopfield Networks.